# DM IMG & LAB

St. George's University

# Preparation



## General Information

The images listed below will assist you in preparing for the lab during open hours. Use these and compare them with the images provided in this document. Not all students will start with station 1 so be sure that you are able to answer the questions of all stations before coming to the scheduled lab session.

### Literature

Available in the *McMinn's & Abrahams' Clinical Atlas of Human Anatomy*:

**Chapter 5: Abdomen and Pelvis**

228A&B , 229, 231, 232A&B, 233A&B, 237, 239, 240A&B, 242, 243, 244, 245A&B, 248, 249, 255, 257A, 262C-F

### Literature

Available in the Gray's Clinical dissector

**Chapter 11: Peritoneal Cavity pg. 186-197**

11.6, 11.10, 11.11, 11.15, 11.17, 11.20, 11.21, 11.22, 11.23, 11.25, 11.26, 11.27

**Chapter 12: Gastrointestinal tract pg. 198-222**

12.2, 12.3, 12.4, 12.7, 12.10, 12.14, 12.15, 12.17, 12.19, 12.20, 12.21, 12.23, 12.27, 12.29, 12.30, 12.32, 12.36, 12.38, 12.39, 12.44, 12.47, 12.48, 12.50, 12.51, 12.52, 12.53, 12.56

# UPPER GI



- Examine the esophagus, stomach, duodenum and pancreas
- Examine the esophageal hiatus, discuss why food does not regurgitate into the thoracic portion during inhalation
- Identify the greater and lesser curvatures, cardiac, pyloric, fundus and body of the stomach
    - Discuss the relationship of the stomach to the liver, spleen, transverse colon, pancreas and kidney
    - Discuss and identify the boundaries of the omental bursa, what is the clinical relevance of this space?
- Describe the structures in the anterior and posterior margins of the foramen of Winslow and discuss its clinical importance
- Examine the gastroduodenal junction and discuss how chyme is prevented from returning to the stomach
- Describe the $1^{st}$, $2^{nd}$ and $3^{rd}$ parts of the duodenum and study its relationship to the pancreas and kidney
- Examine the internal aspect of the duodenum and identify the major duodenal papilla

- Examine the pancreas, identify the; head, neck, body, tail and uncinate process and their relationship to the various surrounding structures.
- Discuss why an ulcer in the posterior wall of the stomach will damage the pancreas.

- Examine the blood supply to the organs described above, discuss the significance of the dual origin of the blood supply to the pancreas.

# UPPER GI



1. Can you Identify the abnormality?

2. Describe the symptoms that you would expect the patient to present with.

https://radiopaedia.org/cases/ad29ecf8ef9a88bd3dc3817a54e7d71a/play?lang=us

# UPPER GI AND GI GLANDS



1. Describe the abnormality seen.
2. How was this image obtained?
3. Describe the anatomical location of the defect/abnormality. Account for its presence using your knowledge of anatomy.
4. Describe the other locations of portacaval anastomosis

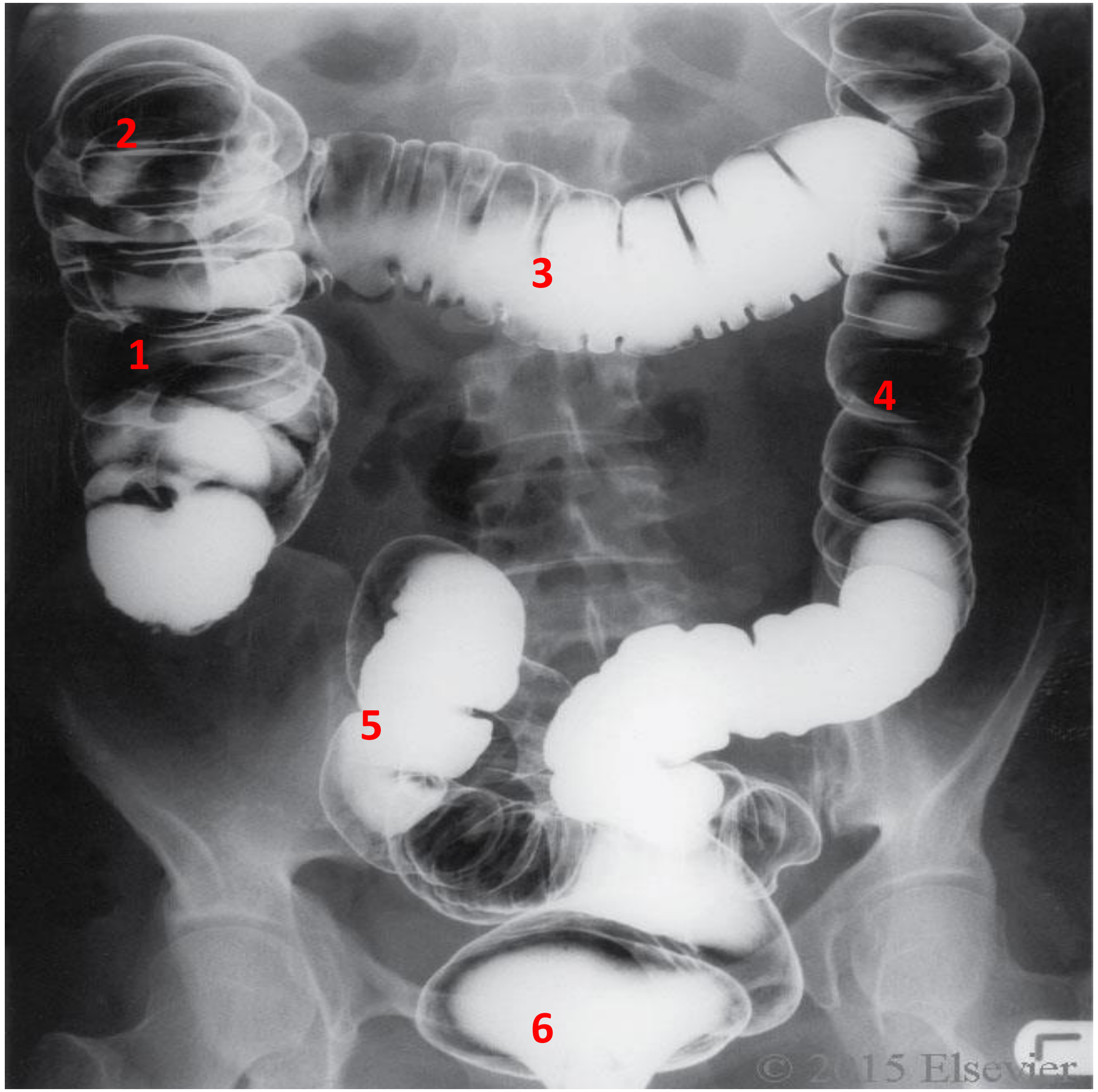
# LOWER GI



- Identify the jejunum, ileum, cecum, appendix, ascending colon, transverse colon, descending colon, sigmoid colon and rectum
- Identify the left and right colic flexure and their relationships
- Recall the location of Mc Burney's point and examine its relationship to the cecum and appendix
- Identify the haustra, tenia coli, omental appendices, left and right colic flexures
- Examine the internal surface of the jejunum, ileum and colon

# LOWER GI

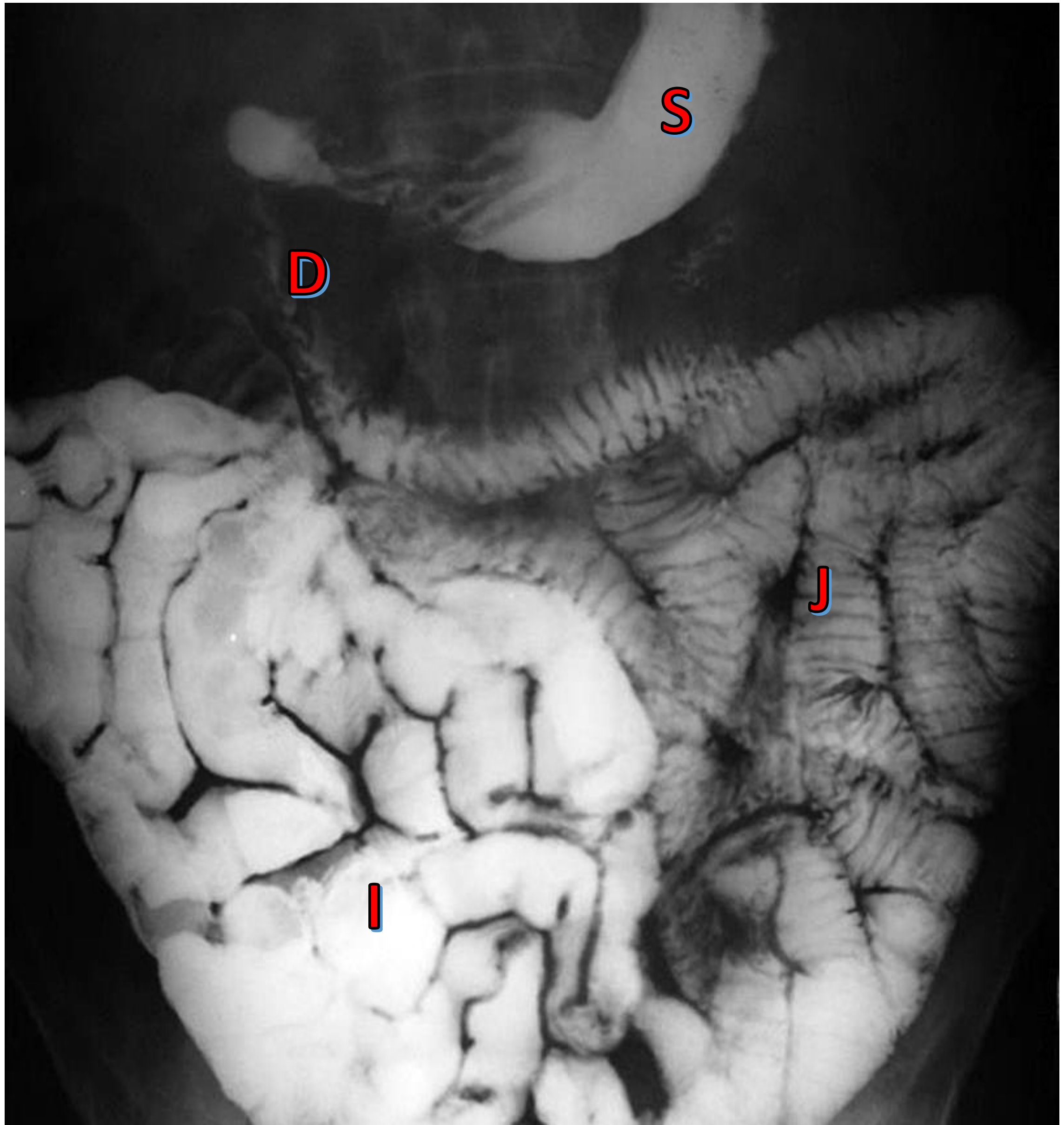





1. What type of imaging modality was used?
2. Which part of the gastrointestinal system is shown in these images?
3. Identify the structures labeled 1-6.

# LOWER GI






1. Identify the stomach, duodenum, jejunum and ileum

2. What is the reason for the feathery appearance of the jejunum?

# PERITONEUM, LYMPHATIC DRAINAGE AND CLINICAL SIGNIFICANCE



- Identify the lesser and greater omentum, mesenteries of the small intestine, greater and lesser peritoneal sacs
    - Discuss where fluid would accumulate in a supine vs standing position
- Identify the paracolic gutters and consider the importance of these spaces
    - Discuss the difference between retroperitoneal and intraperitoneal organs and identify the areas of transition
- Discuss the lymph drainage of each of the parts of the GI track, Discuss the innervation of each portion of the GI track and their relationship to the embryology of the GI

# DIGESTIVE BLOOD SUPPLY



- Identify the celiac trunk, left gastric, splenic, common hepatic, right gastric, proper hepatic, gastroduodenal and cystic arteries.
- Trace the splenic artery and discuss its distribution and identify the gastroepiploic (gastromental) arteries.

  Describe the vessels that form an anastomosis around the stomach

  Discuss the pancreaticoduodenal arteries and the clinical significance of their anastomosis

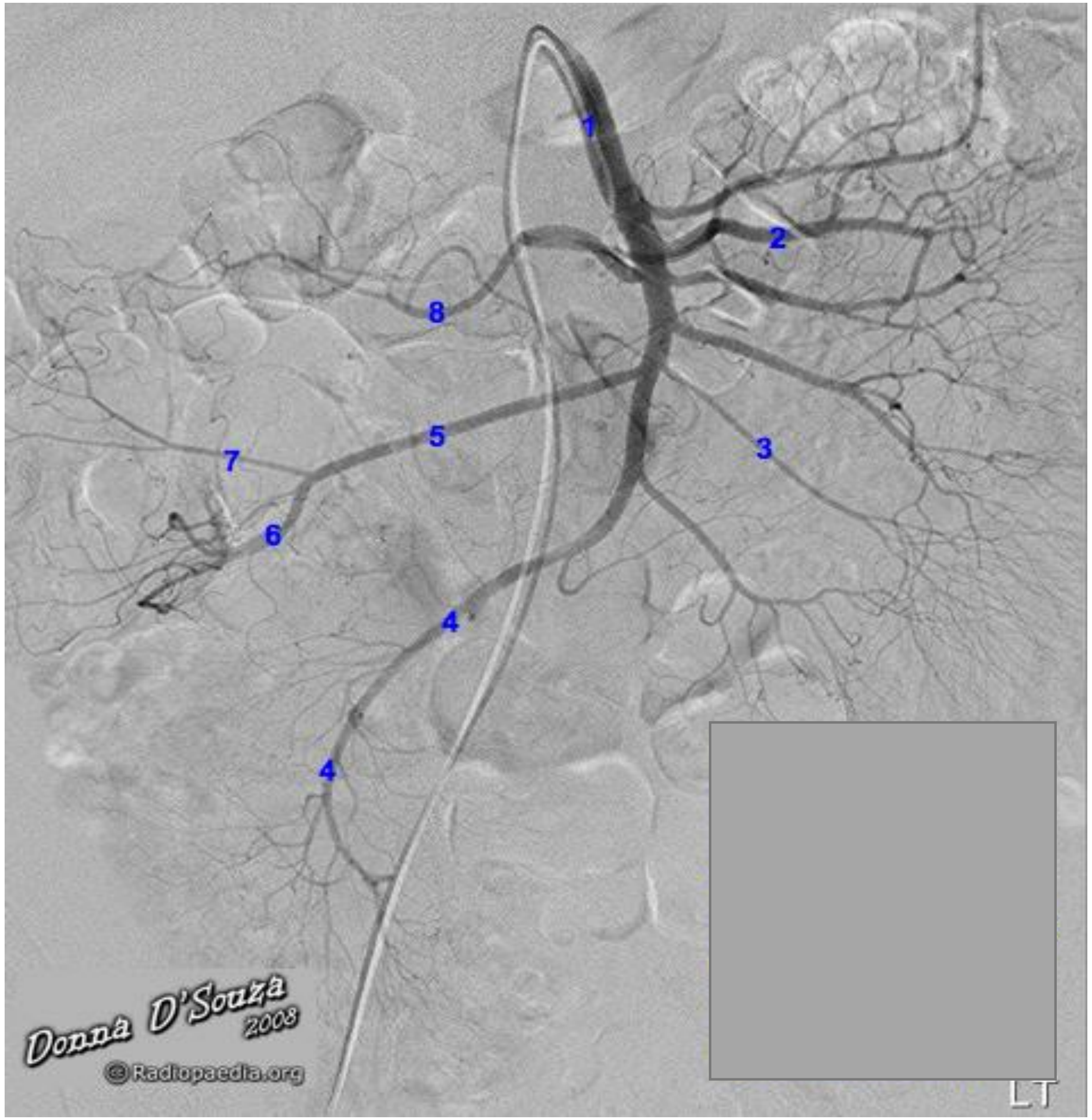
- Identify the superior and inferior mesenteric arteries and their branches
- Examine the blood vessels to the small intestine and describe the differences between the jejunum and ileum (based on vasa recta and arcades)

  Discuss the clinical significance of the marginal artery

  Describe what is meant with a watershed area and discuss its clinical significance

# DIGESTIVE BLOOD SUPPLY

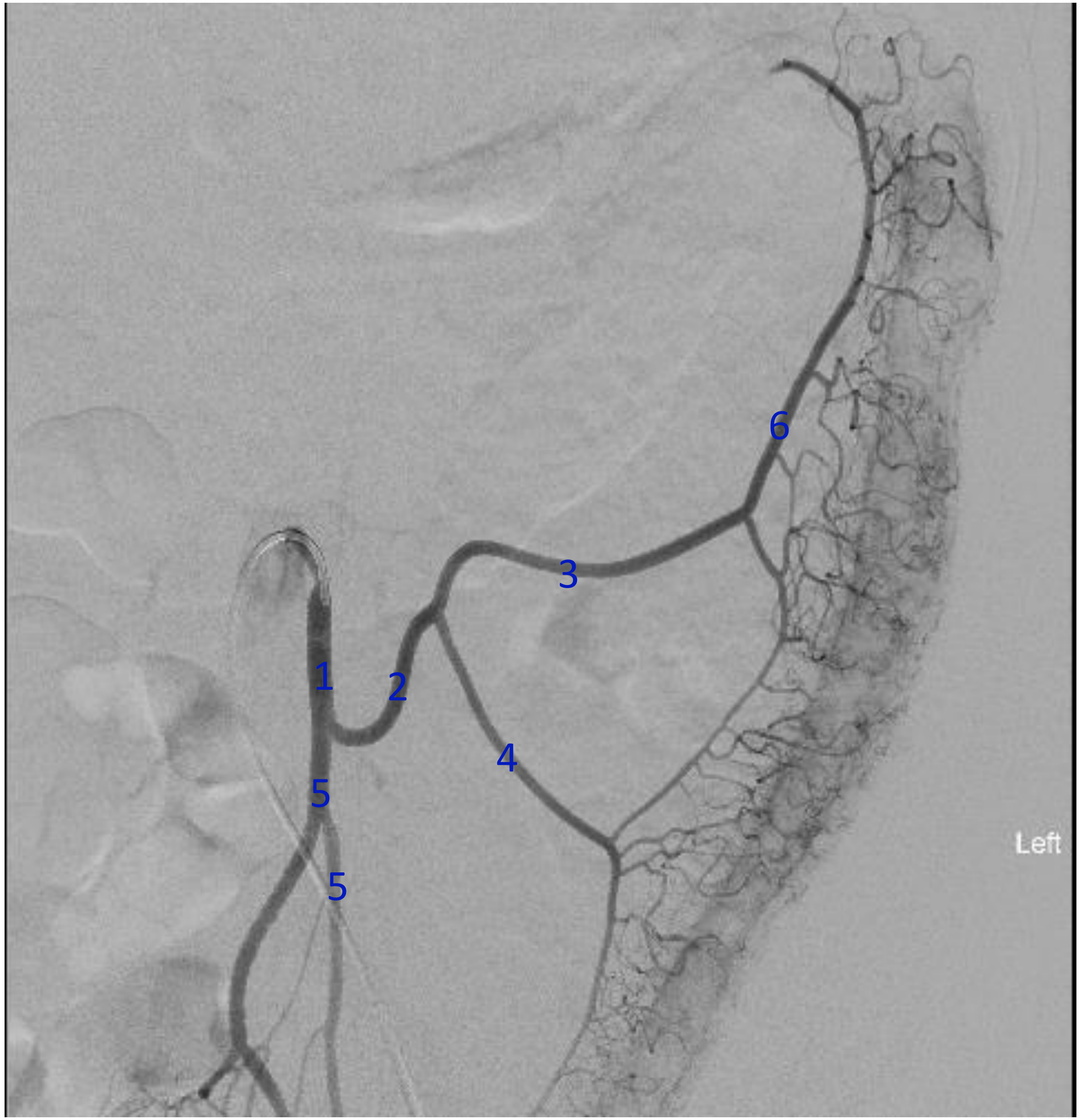





1. Label all the branches of the superior mesenteric artery
2. At which vertebral level does the artery originate

# DIGESTIVE BLOOD SUPPLY






1. Label all the branches of the inferior mesenteric artery
2. At which vertebral level does the artery originate

# ACCESORY DIGESTIVE ORGANS



- Identify the lobes of the liver
    - Discuss the significance between the anatomical and physiological divisions of the liver

- Identify the porta hepatis and the structures that pass through it
- Identify the order and relationships of the structures entering the liver
    - Discuss the blood supply to the liver and identify the vessels
        - What is meant by dual blood supply to the liver?
    - Discuss the venous drainage of the liver

- Identify the gallbladder and its ductal system

    - Discuss the clinical significance of intra-hepatic obstruction vs extra-hepatic obstruction of the ductal system

# ACCESORY DIGESTIVE ORGANS



1. Identify the liver, superior mesenteric vein, splenic vein, portal vein, spleen, splenic artery, small intestine

2. Discuss the formation of the portal vein